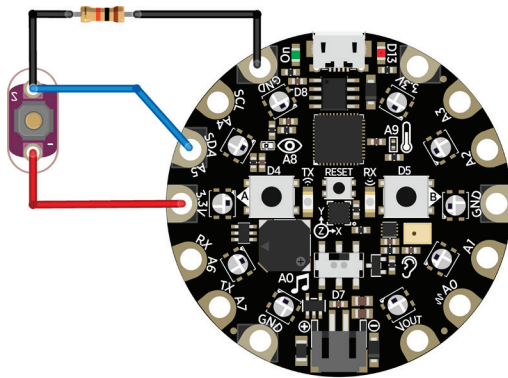
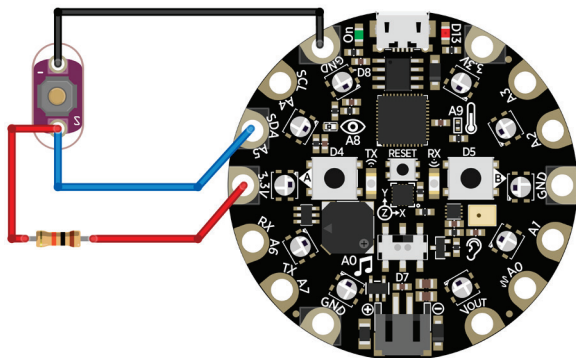
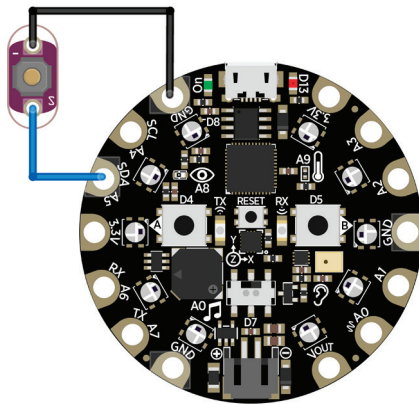




Circuit diagram of CPX with switch and pull-down resistor



Circuit diagram of CPX with switch and pull-up resistor



Circuit diagram of CPX with switch wired for use with internal pull-up resistor; you can also use two alligator clips without the button and simply connect and disconnect the exposed clips at the loose ends.

If the switch is connected to power, you can use a pull-down resistor connected from the digital input pin to ground. If the switch is connected to ground, use a pull-up resistor connected to power. Within this context, something in the range of a 10K Ω resistor will usually do the trick.

If you want to reduce the amount of wiring you have to do, the LilyPad's ATmega chip has an internal pull-up resistor on the digital pins you can activate with the command `pinMode(pinNumber, INPUT_PULLUP)`.

Whichever method you choose, these will all allow the value of the switch to be read by a digital input pin on the microcontroller.

For this example, let's use the circuit with the internal pull-up resistor.

THE CODE

Now that you have a switch connected, how can you write a program that can tell what the switch is doing? Here are a few more Arduino commands that will help you read the value of a digital input:

`pinMode(pin, mode)`

This is something you encountered earlier with digital output. Generally speaking, with digital input, you would set the mode to **INPUT**. This will work with circuit examples that use external pull-down or pull-up resistors. However, if you would like to use the *internal* pull-up resistor, then set the mode to **INPUT_PULLUP**.